

Experiment 1:

RECOGNITION OF STRINGS USING FINITE AUTOMATA

Aim: Implementation of DFA to Accept the following Strings over {a,b}.

- a) accept all strings that ends with 'abb'.
- b) accept all strings that begins with 'aa'.
- c) accept all strings that begins with 'a' and ends with 'b'.
- d) accept all strings that contain the substring 'bab'

Design (Draw transition diagram):

Algorithm (Write algorithm):

Program:

- a) accept all strings that ends with 'abb'.

```
#include <stdio.h>
#include <string.h>

int main()
{
    char str[100];
    int state = 0;
    int i;

    printf("Enter a string (only a and b): ");
    scanf("%s", str);

    for(i = 0; str[i] != '\0'; i++)
    {
        switch(state)
        {
            case 0:
                if(str[i] == 'a')
                    state = 1;
                else if(str[i] == 'b')
                    state = 0;
                else
                {

```

```

        printf("Invalid Input!\n");
        return 0;
    }
    break;

case 1:
    if(str[i] == 'a')
        state = 1;
    else if(str[i] == 'b')
        state = 2;
    else
    {
        printf("Invalid Input!\n");
        return 0;
    }
    break;

case 2:
    if(str[i] == 'a')
        state = 1;
    else if(str[i] == 'b')
        state = 3;
    else
    {
        printf("Invalid Input!\n");
        return 0;
    }
    break;

case 3:
    if(str[i] == 'a')
        state = 1;
    else if(str[i] == 'b')
        state = 0;
    else
    {
        printf("Invalid Input!\n");
        return 0;
    }
    break;
}
}

if(state == 3)
    printf("String Accepted\n");
else
    printf("String Rejected\n");

```

```
    return 0;  
}
```

Output:

Result: